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Nationwide Innovation League

TinyFish
Powering AI Agents for Real World

Project Title

SARATHI

AI - PoweredDigital Business Copilot For Businesses

From local Shops to Smarter Growth



Problem

Small and medium businesses struggle to access AI tools, real-time market insights, and expert digital guidance needed for smarter decisions and sustainable growth.

Our Solution

An AI-powered business assistant that delivers real-time market insights, government scheme information, and smart recommendations to help businesses make better decisions and grow faster.

Smarter Insight | Better Decision | Strong Businesses

Domain

Software Development

Team Details

Team Leader
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Computer Science & Engineering
ABES Institute of Technology, Ghaziabad

Team Member
Satyendra Kumar
Computer Science & Engineering
ABES Institute of Technology, Ghaziabad

Team Name :
Innovators

Our Goal

To build an AI - Powered business assistant that help businesses access real time market insight, government schemes and expert guidance - powered by TinyFish

Smarter Business
Brighter Future

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OBJECTIVE: AI - Powered Digital Business Copilot For Businesses

AIM: To develop an AI-based platform that helps rural entrepreneurs choose suitable businesses, analyse local market conditions, assess financial feasibility, and find relevant government schemes.

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SOLUTION APPROACH

How it Addresses the Problem

A unique system that tracks government capital losses:
Explains theunderlying causes of potential business failure, and predicts future scenarios based on changing market and business conditions.

Easy Usecase
Designedinaway that don't require to enter much detail to use the application. Completely speech based use.

AI-Driven Decision, financial & Government Support: With all the user data and gathered information

SWOT analysis: With all the parameters gathered it & analyses the strength and weakness of the system.

Financial&Government Support: A complete analysis of schemes.

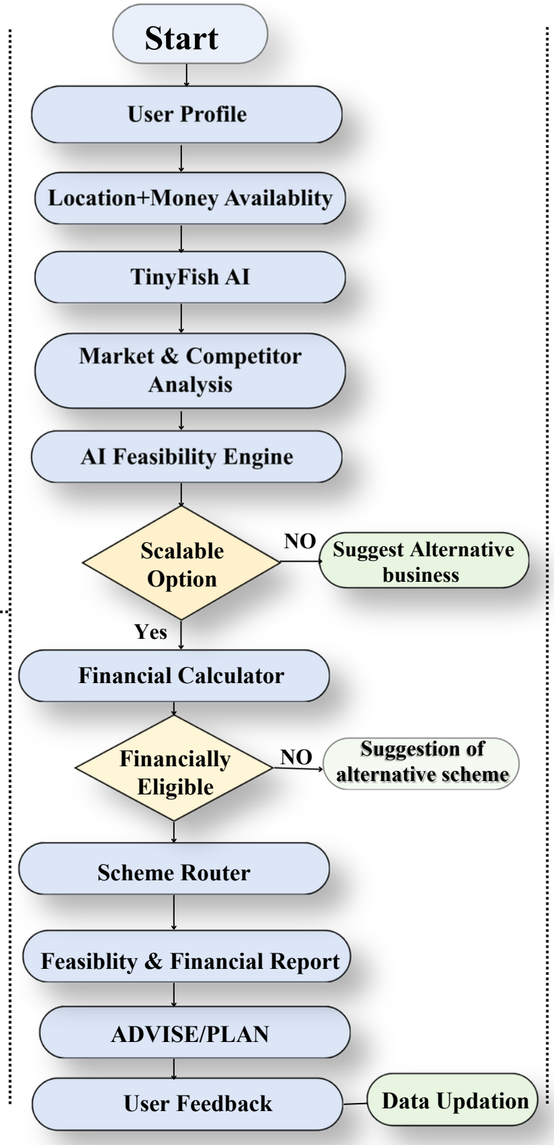
Things to Target

Rupees 30 LakhCrore
Estimated MSME Credit gap in India

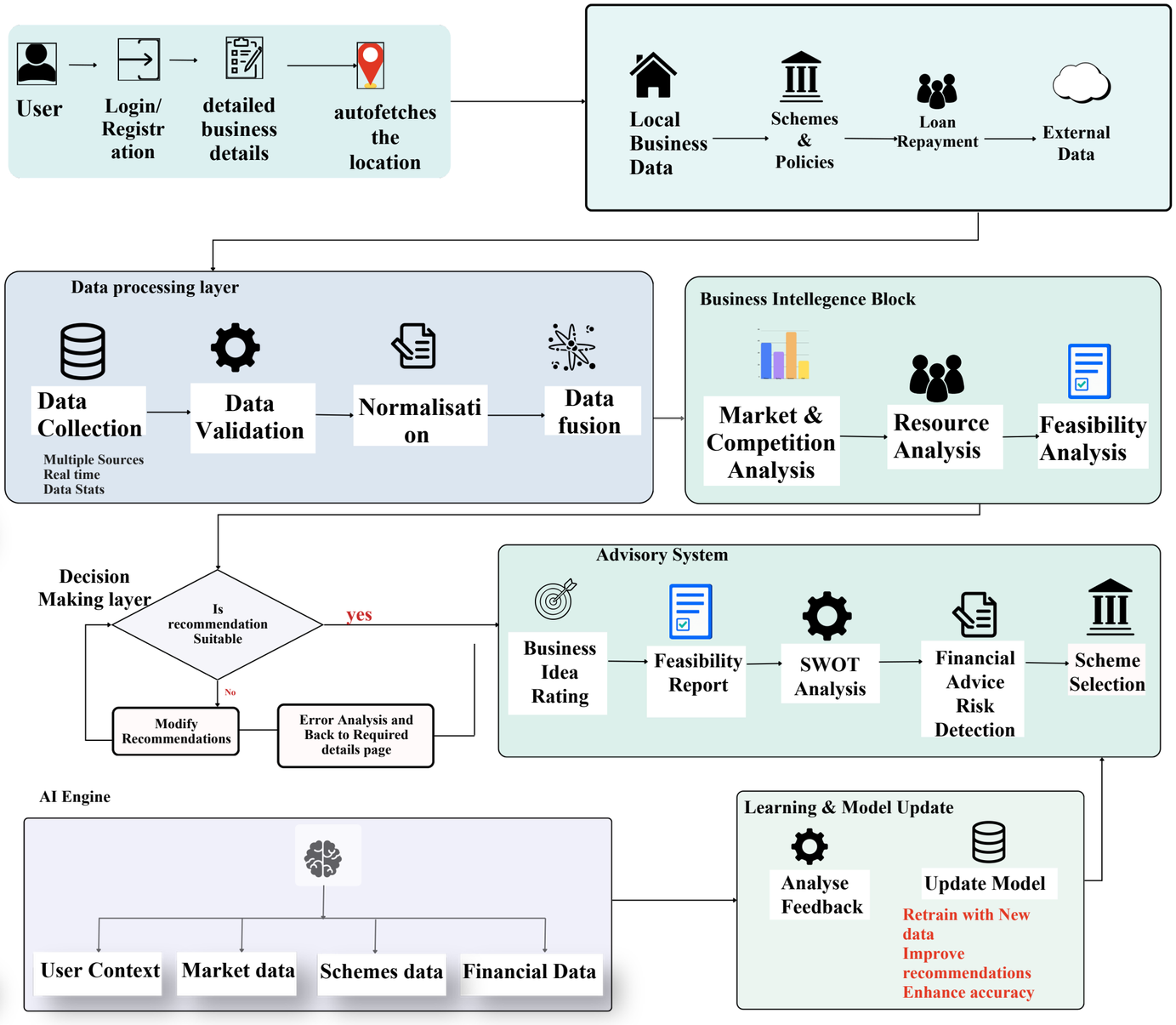
3.97 Crore rural unincorporated establishments in year 2023-2024

17% of rural people use informal way of borrowing money

In 2022-2023 rural enterprsises grown from 3.56 crore to 4.24 crore in 2025 resulting in 19.1% growth



WORKFLOW



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Problem & Detailed Explanation

DETAILED EXPLANATION

Overview

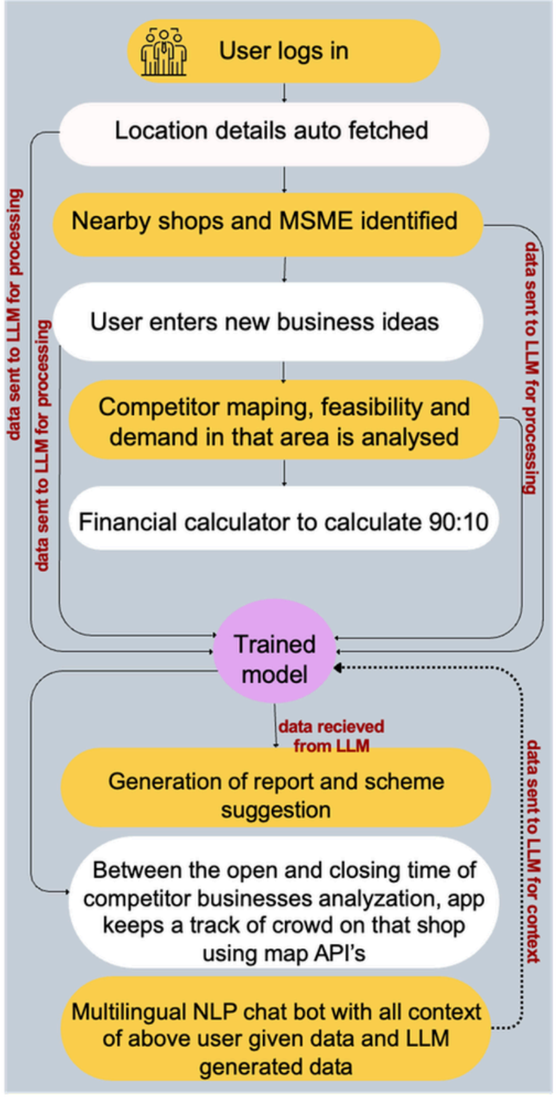
Our multilingual AI chatbot acts as a personal business advisor for rural entrepreneurs. It understands the user's business idea, location, budget, skills and experience, then combines this with local market and financial data to check business feasibility, identify suitable government schemes, and provide personalized recommendations and next steps—all through a simple chat-based interface.

Technologies Used

Utilizes natural language processing (NLP) and machine learning to provide accurate information on business ideas, uses self trained model to filter and generate reports, **scheme suggestion**, API's to **gather MSME and nearby businesses**. data **collection from public listing, map API's and**

Key Features

- **Personalized Business Matching –** Matches business opportunities with the user's budget, skills, experience and location.
- **Local Competition & Demand Analysis** – Uses nearby-businesses and MSME data to identify market gaps and competition.
- **Business Failure Insights** – Identifies potential risk factors and reasons a proposed business may fail before investment.
- **Intuitive User Interface**: Designed for ease of use for rural users, accommodating all technical skill levels.
- **Instant Information Access:** Users can inquire about businesses, opportunities and feasibility
- **Continuous Context** – Retains the entrepreneur's business context across conversations for more relevant advice.
- **Threat Assessment and Competition Mapping**
- **Loan Eligibility and Scheme Matching**
- **Feasibility report and graphical representation of data**



HOW IT ADDRESSES THE PROBLEM

- **Personalized Business Guidance:** Uses the entrepreneur's budget, skills, experience, location, and business idea to provide relevant guidance.
- **Business Feasibility Assessment:** Evaluates market, operational, and financial feasibility to reduce the risk of choosing an unsuitable business.
- **Local Market Intelligence:** Uses MSME and nearby-business data to identify local demand, competition, and potential market gaps.
- **Simplified Scheme Access:** Matches entrepreneurs with relevant government loans, subsidies, and schemes based on their business requirements.
- **Multilingual Chat Interaction:** Enables users to communicate naturally in their preferred language through a simple chatbot.
- **AI-Based Decision Support:** Combines user context and multiple data sources to provide business recommendations, risk insights, and actionable next steps.
- Locality, population, and business type and available capital.
- The application makes a business model and revenue model depending upon capital use and population of the area and competition ***[risk & Failure Prediction, Profitability & Financial Projection]***

INNOVATION AND UNIQUENESS OF SOLUTION

Innovative Approach: The model with integrated API's keeps track of competitor businesses and finds out the market businesses captures by algorithm (crowd, working timing of business, photos if available on maps)

Data like photos, rating of businesses and working hours are used in finding out the scale of competitor then this data is used in report generation, and chat section too. competitor mapping is done with same method.

The chat uses your data along with nearby business and local data from Maps, APIs, and users to give practical advice based on what's actually happening around you.

This way of gathering data and processing leads us to understand why businesses fail or do not grow. It also helps track where the invested capital and government support are being lost.The system records the actual reasons behind business failure.
It identifies what went wrong and what can be done to correct it. so we have the only available system that This creates a system that keeps track of business failures and helps reduce future loss of funds.

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How we Plan Integrate the TinyFish API

Real-time Web Intelligence | Live Data | Actionable Insights

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Objective

Enable SARATHI to go beyond static business data and provide live, actionable information from the real web whenever a small-business owner needs information that changes over time.

Where TinyFish Fits

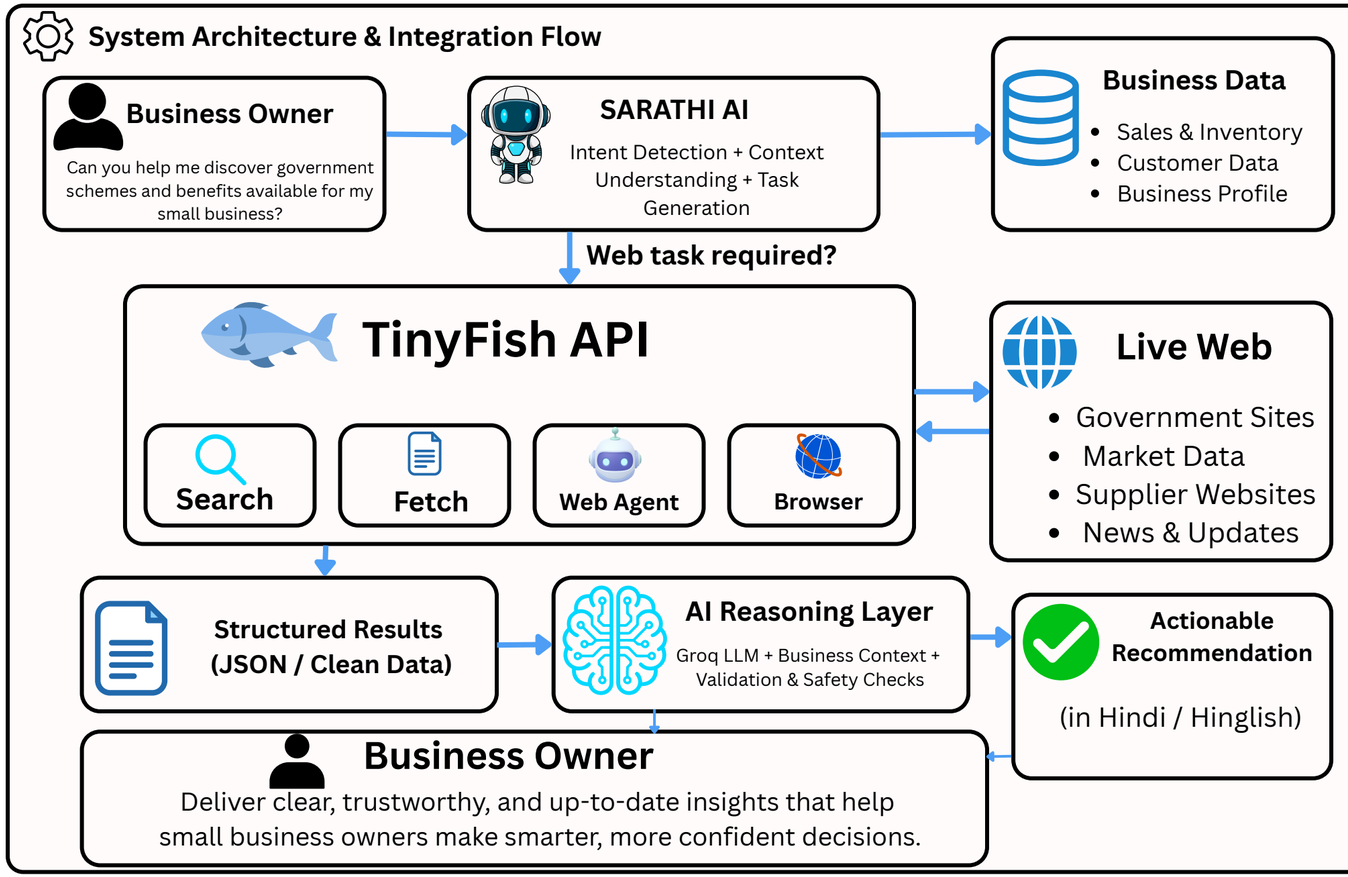
TinyFish acts as the Web Intelligence & Automation Layer of SARATHI. Instead of training our model on constantly changing information, SARATHI delegates web-based tasks to TinyFish and uses the returned results as fresh context.

TinyFish

Web Agent | Search | Fetch | Browser

AI agents that search, extract, browse and act on live web information.

System Architecture & Integration Flow



Key Use Cases

1. Government Scheme Discovery

Which government schemes are available for my business, and how can I apply for them?

2. Live Market & Price Intelligence

What is the current market price of my product today?

3. Supplier & Product Research

Find reliable suppliers for this product and compare their prices, availability, and credibility.

4. Business Opportunity Monitoring

What new business opportunities, local programs, and initiatives are currently available for my business?

Reliability & Safety Layer

- Schema Validation
- Source Verification
- Freshness Check
- AI Reasoning
- Official / Credible Sources
- User Confirmation for Critical Actions

Detailed Integration Steps

1. **User Query** — Understand the user's needs, location, and type of business.
2. **Generate Web Task** — Create a goal-based web research plan to find relevant, reliable, and up-to-date information tailored to business needs.
3. **TinyFish Executes** — Search relevant websites, fetch useful content, browse live sources, and extract structured data for accurate, actionable business insights.
4. **Validate & Verify** — Check schema, source, freshness, completeness
5. **AI Reasoning** — Combine web + business data
6. **Final Output** — Present recommendation to user

Why TinyFish is Essential

Without TinyFish	With TinyFish
✗ Outdated information	✓ Fresh information
✗ Manual web searching	✓ Automated web research
✗ No live market intelligence	✓ Dynamic website interaction
✗ Difficult to handle dynamic sites	✓ Structured results
	✓ Real-world actionable intelligence

TinyFish helps SARATHI search the web and find the latest information that small business owners need. It helps them discover government schemes, check current market prices, find trusted suppliers, and explore new business opportunities. Instead of spending hours searching different websites, business owners can get useful information in one place and make better decisions for their business.

Business Owner → SARATHI AI → TinyFish Agent → Real-world Web → Verified Information → Actionable Recommendation

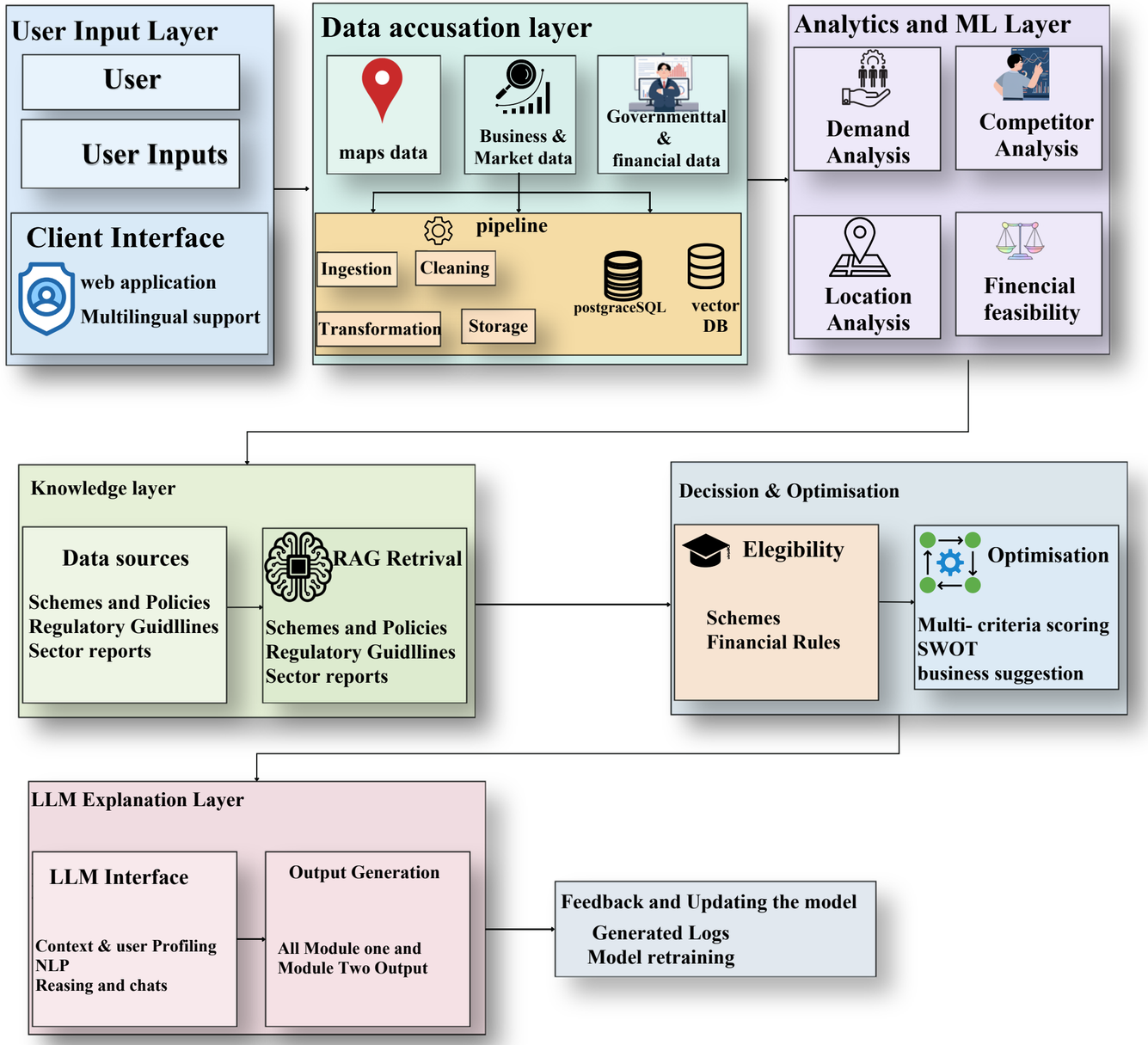
TinyFish

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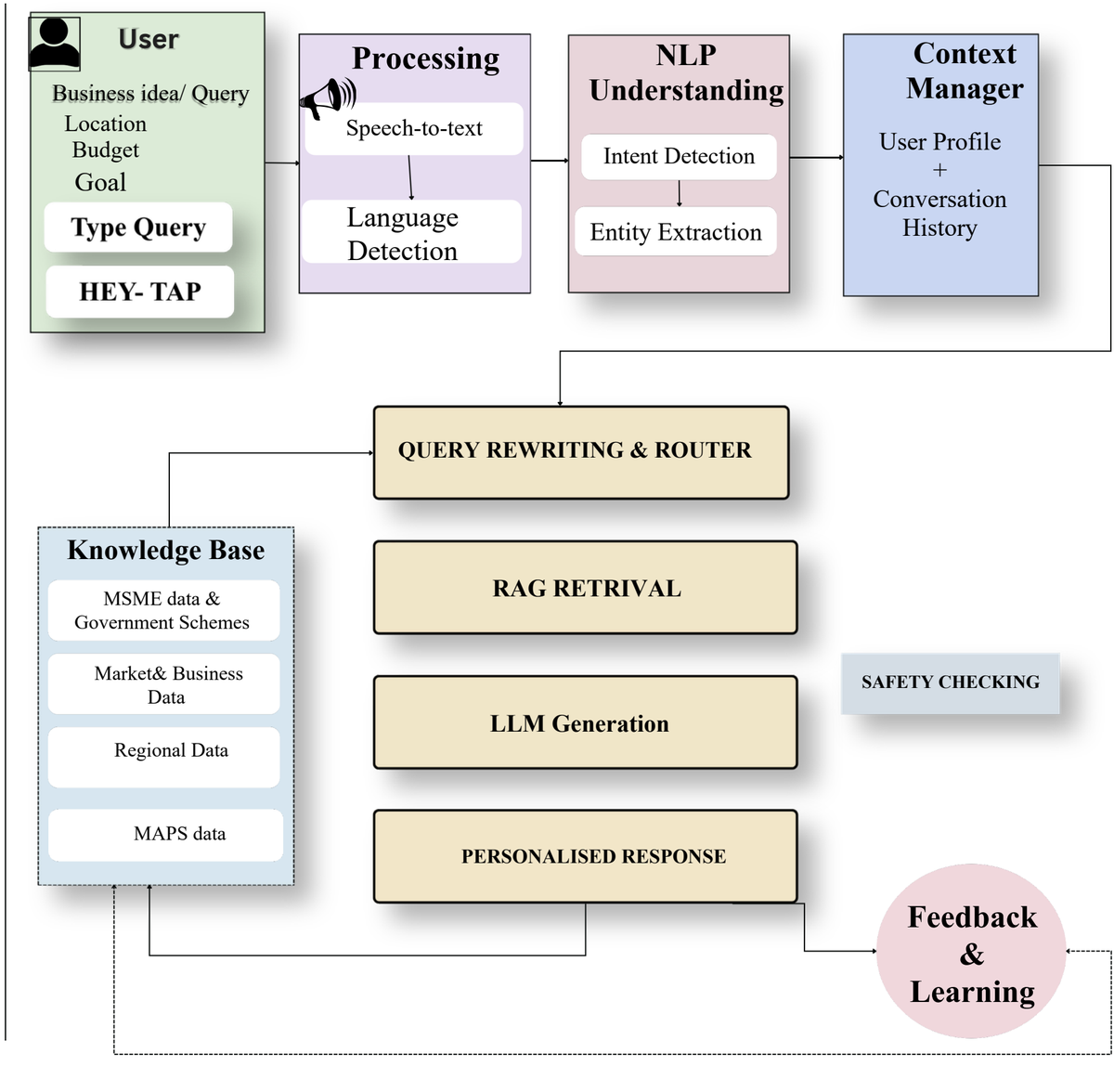
FLOWCHARTS

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TECHNICAL FLOW CHART



RAG ENABLED CHAT SYSTEM



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FEATURES & DATA FLOW DIAGRAM

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USE CASE

1. Rural Entrepreneur

2. Existing Local Businesses

3. Government Departments

. Banks / Financial Institutions

- User Registration & Business Input Processing
- Data Collection
- Competitor Discovery & Mapping
- Demand and Location Analysis
- Cost, Revenue & Profitability Estimation
- Risk & Failure Prediction
- Government Scheme Matching & Fund Tracking
- Personalized Business Plan & Recommendations
- Multilingual AI Chatbot
- Continuous Learning

- Business Insights for local Business
- Policy& Planing for gov.
- Credit Asesment for Banks
- System of report analysis ans Administration

DATA FLOW DIAGRAM

- **User Input**
- **Location&AreaContext**
  Roads and Connectivity
  Locality & Benificary
  Jones
- **Places API**
  Nearby Business
  Category Location
  Opening Hours
  Pricing, Rating, business Status, Photos
- **Competitor Details**
  Popular Times
  Area Busyness Trends
  Aggregated & Anonymised Data
- **Govt& Scheme data**
  Schemes, ElegibilityCriteria
  Subsidy Loans
  Funds
  Policy Explanation
- **Financial Data**
  Rent, Equipment Cost
  Utilities
  Working Capital and requirement

API/ Data Collection → Normalisation & Data Cleaning → Compititor maping → Competitor Profile Details

Processed & structured data

Demnad & business Analysis → Location & Oppurtunity Analysis → Business Scale Analysis → Financial feasibility Engine

Competitor Gap & Advantage Analysis | SWOT Analysis | Funds & Govt. Schemes Matching

AI analysis and Decission Making

Model Retraining

Required Output | NLP based Chat system

TECHNOLOGY STACK

FRONTEND: HTML, CSS, React

BACKEND: Flask

DATABASE

Map: OpenStreetMap

AI/NLP FRAMEWORK: RAG

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IMPACTS & BENIFITS

IMPACTS

- Stronger Economy
- Easy interpreneurship
- Reduced Business Failure Rate
- Better Utilization of Government Funds
- Reduced Capital Misallocation
- Daily situation based resources availability insights

BENIFITS

- Early Risk Identification
- Data-Driven Business Selection
- Faster Course Correction
- Better Financial Planning
- Personalized Business Guidance

UNIQUENESS

A complete voice enabled systemtobeusedeasilybyvillagersgets realtime data and realtime insights of local business.

Map based insights for user to inderstand the market and strength of competitors

By collecting and analyzing business data, the system identifies why businesses fail or stop growing, tracks where invested and government funds are lost, and records the actual causes of failure. This helps to prevent similar failures and reduce future financial losses.

Innovative Approach: Integrated multipleple APIs to track competitors and predict their market share using crowd levels, business hours, and available map photos of their business.

Business data helps assess competitor scale and supports reports, chat, and competitor mapping (data from images on maps and other derails)

Challenges Addressed

- Hard to judge demand and real competition
- No data of competitors and their strength to local men in village
- Difficult to estimate risk
- No idea of Resource availibility in different climate and situations
- Financial and funding assessment
- Language barrier
- Lack of monitoring and failure

Overall Viability

- Solves a real and relevant problem for enterprenures
- Technically and operationally feasible with
- available resources
- Financially sustainable as generates
- revenue opportunities
- Scalable to many regions and business
- Requires no fee to operate
- Scalable as it also works in analysing
- resources and opportunities

FEASIBILITY

Analysis of feasibility

Growing Demand: There is a growing demand of advisory system in rural areas. the agencies cost huge sum of money, an alternate system is required

Technological Availability: Access to advisory system where the costs are too low and every introduced feature is possible to make

Data Feasibility: Data from Govt. site and maps are enough as they give full insights of all local businesses

Scalability: Modular design and Support multiple Languages. District wise data and intelligence

UserEngagement: Easy to use, Hey-Tap, Rag enabled chat system

API Integration (Technical feasibility and Technical Viability)

- Multi Platform Api
- Government data Filtering and finding out all the local competitors

API:

- ~~Open street map + overpassAPI~~
- ~~Google places Api~~
- ~~Mappl API~~
- ~~data.gov.in~~
- ~~Udham/ MSME data sets~~

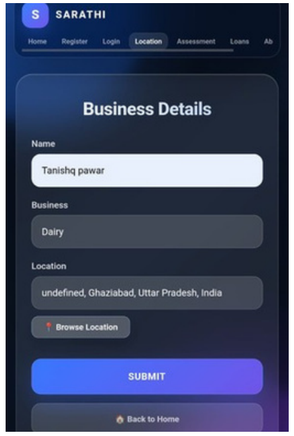
Strategies to overcome these problems

- **User collective data in arranged order**
- **Gather data from multiple sources to get all local business**
- **AI based Feasibility and Analysis**
- **Early warnings**
- **Resources and Its availability**

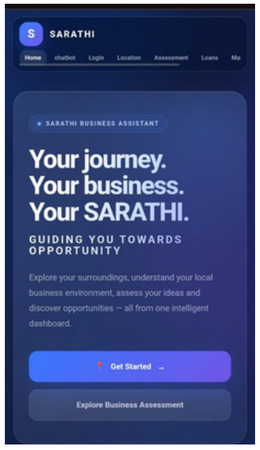
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RESEARCH & REFERENCES

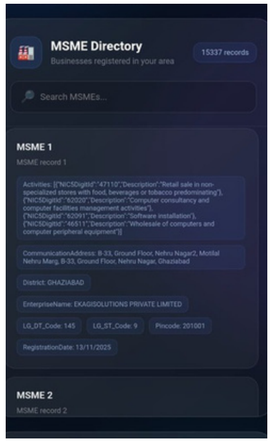
PROTOTYPE



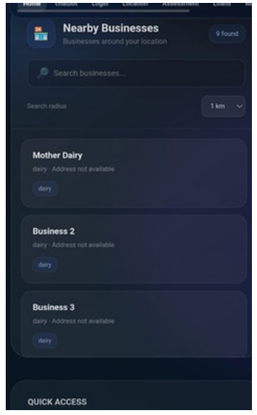
Business Details



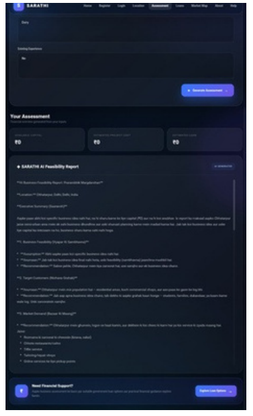
Home page



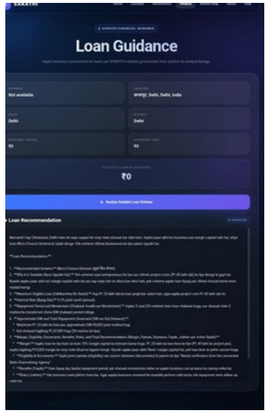
Local MSME details



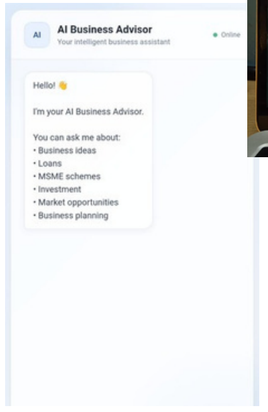
Local Martket details with competitor size, markety and capital details



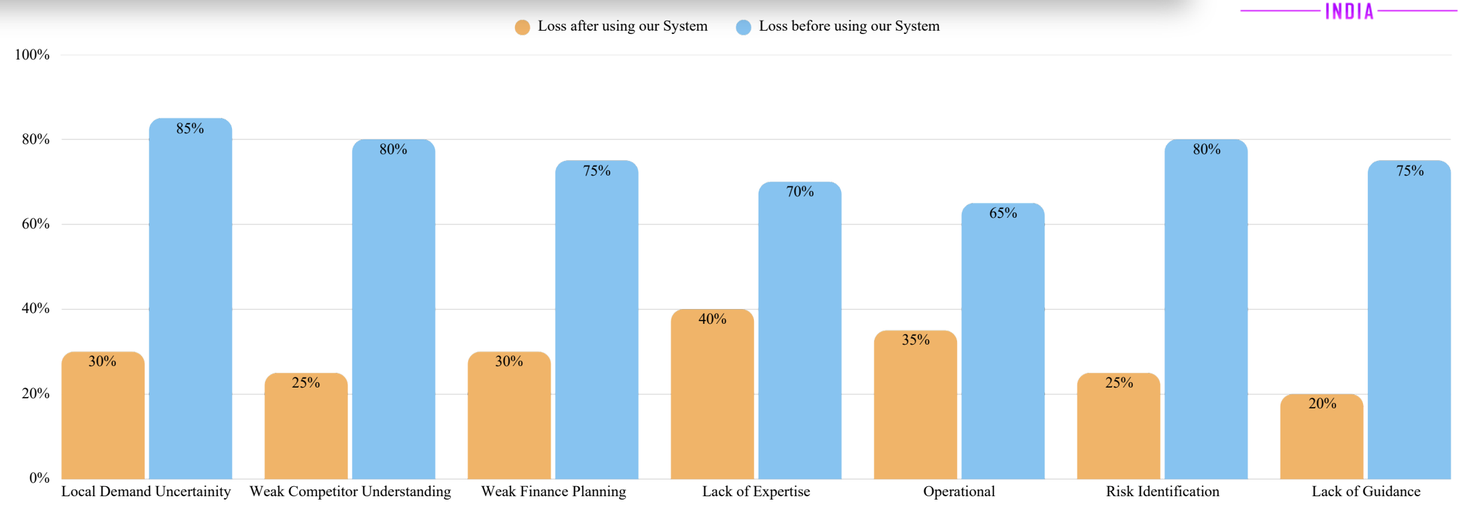
NLP based chat system
and one tap
hey cloud system

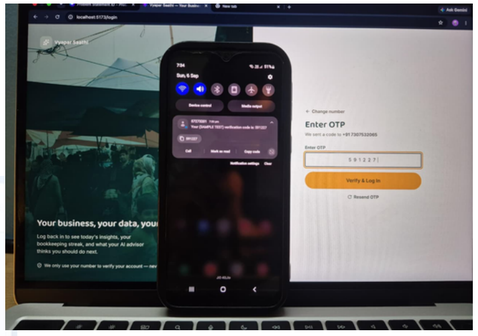


Feasibility Report

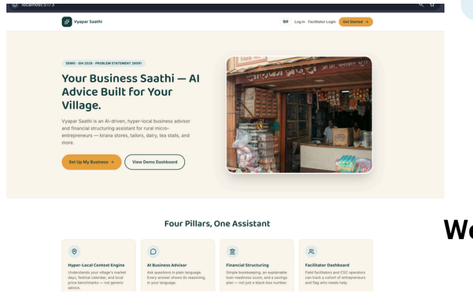


Govt. scheme and Loan assesment





Authentication



Web Home page

RESEARCH

Title: Location Based Business Recommendation Using Spatial Demand
DOI: 10.3390/su12104124

Title: Optimal Retail Location: Empirical Methodology and Application to Practice
DOI: https://doi.org/10.1287/msom.2018.0759

Title: Location Intelligence: An Innovative Approach to Business Location Decision-making
DOI: https://doi.org/10.1111/j.1467-9671.2011.01253.x

Title: Financial Access and Entrepreneurship by Gender: Evidence from Rural India
DOI: https://doi.org/10.1007/s11187-024-00925-z

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- ONDC
- Udyam Portal
  Google maps